5th Semester Questions 18_19

Answer any 02 out of 03 Questions

1 Resolve the following system using the Cramer's Rule. 7.5

$$0.14\,X_1 - 0.1X_2 - 0.2X_3 = 7.85$$
$$0.10X_1 + 7X_2 - 0.3X_3 = -19.3$$
$$0.30X_1 - 0.2X_2 + 10X_3 = 71.4$$

2 Use the Gauss-Elimination technique to resolve the following system. 7.5

$$3X_1 - 0.1X_2 - 0.2X_3 = 7.85$$
$$0.10X_1 + 7X_2 - 0.3X_3 = -19.3$$
$$X_1 - 0.2X_2 + 10X_3 = 71.4$$

3 Apply the Factorization process to locate the root of the following system 7.5

$$X_1 + X_2 - X_3 = 2$$
$$2X_1 + 3X_2 + 5X_3 = -3$$
$$3X_1 + 2X_2 - 3X_3 = 6$$

Dept. of Computer and Communication Engineering

Mid: I   Semester: 5th   Batch : 16th   Marks 15   Session 2018-2019

Course Code: CCE 312   Time: 1 hour   Course Title: Numerical Method

1. A total of 8,500 taka was invested in three interest earning accounts. The interest rates were 2%, 3% and 6% if the total simple interest for one year was 380taka and the amount invested at 6% was equal to the sum of the amounts in the other two accounts, then how much was invested in each account? (use Cramer's rule).Cramer's Rule with python code.

2. Use the Gauss-Jordan technique to solve the same system as in

$$3x1 - 0.1x2 - 0.2x3 = 7.85$$
$$0.1x1 + 7x2 - 0.3x3 = -19.3$$
$$0.3x1 - 0.2x2 + 10x3 = 71.4$$

3. Use Gauss elimination to solve:

$$8x_1 + 2x_2 - 2x_3 = 8$$
$$10x_1 + 2x_2 + 4x_3 = 16$$
$$12x_1 + 2x_2 + 2x_3 = 16$$

Employ partial pivoting, and check your answers by substituting them into the original equations.

4. *Three masses are suspended vertically by a series of identical springs where mass 1 is at the top and mass 3 is at the bottom.*

If $g = 9.81\ m/s^2$, $m1 = 2$ kg, $m2 = 3$ kg, $m3 = 2.5$ kg, and the k's = 10 kg/s, *solve for the displacements x.*

5. *Solve the following systems of linear equations by* Gaussian *elimination method*

$$2x - 2y + 3z = 2,\ x + 2y - z = 3,\ 3x - y + 2z = 1$$

Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

Dept. of Computer and Communication Engineering

Mid: I Semester: 5th Batch : 16th Marks 15 Session 2018-2019

Course Code: CCE 313 Time: 55 Min Course Title: Computer Network

1 An classful address in a block is given as 222.8.17.9. Find the number of addresses in the block, the first address, and the last address. Draw also the block diagram of this IP address topology. 3

2 Suppose you have given a classful block of IP 140.15.0.0. Now you need to divide this IP block to four subnetwork with equal IP address space of each block. Now extracting the first address last address, subnetwork mask to follow the proper procedure and also draw the diagram of the sub network. 5

3 Suppose Alice, who always accesses the Web using Internet Explorer from her home PC, contacts Amazon.com for the first time. Let us also suppose that in the past she has already visited the eBay site. Now, what will happen, when the request comes into the Amazon Web server for the first time and then one week later? Illustrate the communication process between Alice's browser and the Amazon web server with respect to cookies. 4

4 What is the function of conditional GET? 3

Patuakhali Science and Technology University

Department of Computer Science and Information Technology

5th Semester (Level-3, Semester-I), **Midterm Examination** of **B.Sc. Engg.(CSE)**, January-June/2021, Session: 2018-19

Course Code: CIT-311 Course Title: Microprocessor and Assembly Language

Full Marks: 15 Duration: 50 minutes

[Figures in the right margin indicate full marks]

Answer all the following questions.

1. What is a von Neumann machine? Write down the major difference between Intel 8085 and 8086 microprocessor. 2
2. What is the difference between an intersegment and intrasegment jump? Show which JMP instruction assembles (short, near, or far) if the JMP THERE instruction is stored at memory address 10000H and the address of THERE is: 2
   i. 10020H
   ii. 11000H
3. Convert an 8B9E004CH from machine language to assembly language. If a MOV SI,[BX+2] instruction appears in a program, what is its machine language equivalent? 3
   Bx [BP+
4. What will be the CS:IP of physical address BCDEFh where CS=FFF0? How is the local descriptor table addressed in the memory system? 3
5. Which register locates the global descriptor table? Describe the content of the segment register at protected mode memory addressing. 3
6. Explain the instruction with respect to 8086 microprocessor MOV AX, [BX]. 2

FFFDC : FFFBCEF

Patuakhali Science and Technology University

Department of Computer Science and Information Technology

5th Semester (Level-3, Semester-I), **Midterm Examination** of **B.Sc. Engg.(CSE)**, January-June/2021, Session: 2018-19

Course Code: CIT-313 Course Title: Computer Architecture

Full Marks: 15 Duration: 50 minutes

[Figures in the right margin indicate full marks]

Answer all the following questions.

1.	List and briefly define the possible states that define an instruction execution.	3
2.	Consider a hypothetical microprocessor generating a 16-bit address (for example, assume that the program counter and the address registers are 16 bits wide) and having a 16-bit data bus. i) What is the maximum memory address space that the processor can access directly if it is connected to a "16-bit memory"? ii) What is the maximum memory address space that the processor can access directly if it is connected to an "8-bit memory"? iii) What architectural features will allow this microprocessor to access a separate "I/O space"? If an input and an output instruction can specify an 8-bit I/O port number, how many 8-bit I/O ports can the microprocessor support? How many 16-bit I/O ports? Explain.	3
3.	Give the elements for designing bus.	1.5
4.	Why study Computer Organization and architecture? Describe the structure and functions of a computer.	3
5.	Describe the evolution of DRAM and processor characteristics.	2
6.	Prepare a question on **semiconductor main memory** and answer it yourself?	2.5

5th Semester (Level-3, Semester-I), **Midterm Examination** of **B.Sc. Engg.(CSE)**, January-June/2021, Session: 2018-19

Course Code: CIT-315 Course Title: Artificial Intelligence

Full Marks: 15 Duration: 1 hour

1. What is blind search? How to evaluate an algorithm's performance? Compare search strategies in terms of the four evaluation criteria set. 05
2. Write the advantage of informed search. How to make heuristic function? Define and illustrate admissible heuristic and consistent heuristic for estimating optimality of A* search algorithm. 05
3. Define logical agent. Write down a simple algorithm of a generic knowledge-based agent. Given a percept, the agent adds the percept to its knowledge base, asks the knowledge base for the best action, and tells the knowledge base that it has in fact taken that action. 03
4. What is propositional logic? Drives a propositional logic from Wumpus world is a cave consisting of rooms connected by passageways.

Department of Computer and Communication Engineering
Patuakhali Science and Technology University

Final Examination

CCE 310 - Software Development Project 1

Time Duration: 3 Hours. *Full Marks:70*

1. Write a sample project proposal based on your developed project in Software Development Project 1 course. Please try to cover the following part in your sample project proposal. 15

   - Objective and Introduction
   - Proposed Solution
   - Scope of work
   - Schedule and Timeline
   - Authorization and Conclusion

2. Project Work 30
3. Project Report 10
4. VIVA 15

11/03/2022 22:31

Dept. of Computer and Communication Engineering

Faculty of Computer Science and Engineering

Patuakhali Science and Technology University

Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: I Session: 2018-2019

Course Code	**Course Title**	**January-June 2021**	**Credit:** 03
CCE 311	Numerical Methods		**Time:** 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 [A.] Resolve the following system using the Cramer Rule. 7

$$0.40X_1-0.1X_2-0.2X_3=7.85$$
$$0.10X_1+7X_2-0.3X_3=-19.3$$
$$0.30X_1-0.2X_2+10X_3=71.4$$

x, 22.135
y, -21.798
z = 6.919

[B.] Use the Gauss-Elimination technique to resolve the following system. 7

$$0.3X_1-0.1X_2-0.2X_3=7.85$$
$$0.10X_1+7X_2-0.3X_3=-19.3$$
$$X_1-0.2X_2+10X_3=71.4$$

x = 28.027
= -3.974
= 4.277

1.640
533

2 [A.] Apply the Choleski's process to locate the root of the following system 7

$$X_1+X_2-X_3=2$$
$$2X_1+3X_2+5X_3=-3$$
$$3X_1+2X_2-3X_3=6$$

x = 1
y = 0
z = -2

[B.] 7

i. How can we eliminate the error in the Trapezoidal rule by applying the Simpson's rule?

ii. Integrate using Simpson's 3/8 rule.

$f(x)=0.2+25x-200x^2+675x^3-900x^4+400x^5$ from a=0 to b=0.8

3

3 [A.] Solve the following system using the Gauss-Jordan method. 7

$$0.7X_1-0.1X_2-0.2X_3=7.85$$
$$0.10X_1+7X_2-0.3X_3=-19.3$$
$$X_1-0.2X_2+10X_3=71.4$$

~~x = 12.498~~
= -2.685
= 5.836

$2x^2-5x+3=0$

[B.]

i. Show two scenarios. In the case of an iteration process, there is convergence and divergence.

ii. Write down the algorithm of Bisection Method.

4 [A.] Fit a second-order polynomial to the data of the following table. Also find out standard error $S_{y/x}$ 7

x_i	y_i
0	2.1
1	7.7
2	13.6

x = 2.1
5.95
0.15

[B.] Derive equation for linear regression and find out a_0 and a_1. 7

5 [A.] i. State Weddle's Rule. 7

ii. Show that $x_r = x_u - f(x_u)(x_l - x_u)/(f(x_l)-f(x_u))$ in case of false position method.

[B.] Given $dy/dx = ½ (x+y)$, $y(0)=2$, $y(0.5)=2.636$, $y(1.0)=3.595$, $y(1.5)=4.968$. Find $y(2)$ by Milne's method. 7

$y_n = y_0 + \int_{x_0}$

6 [A.] Given $dy/dx=1+xy$ and $y(0)=1$. Calculate $y(0.1)$, $y(0.2)$ using Picard's method. 7

[B.] Use the Euler's method to numerically integrate $dy/dx = -2X^3 + 12X^2 - 20X + 8.5$ from X=0 to X=4.0 with a step size 0.5. The initial condition at X=0 is Y=1. 7

5 - (-2)

Course Code	Course Title	Semester: I Session: 2018-2019	
CCE 312	Numerical Methods Sessional	January-June 2021 Marks: 70	Credit: 0.75 Time: Hr

1. a. Consider finding the root of $f(x) = e^{-x}(3.2 \sin(x) - 0.5 \cos(x))$ on the interval [3, 4], this time with $\varepsilon_{step} = 0.001$, $\varepsilon_{abs} = 0.001$. Show also the result in tabular format and plot the solution using graphically using Bisection method. 35

   b. For the initial value problem (IVP) $y' - y = -\frac{1}{2}e^{\frac{t}{2}}\sin(5t) + 5e^{\frac{t}{2}}\cos s(5t)\ y(0) = 0$ 35

   Use Euler's Method to find the approximation to the solution at t=1, t=2, t=3, t=4, and t=5. Use h=0.1 ,h=0.05, h=0.01, h=0.005h=0.005, and h=0.001h=0.001 for the approximations. Compare them to the exact values of the solution at these points. Show also the result in tabular format and plot the solution data using graphically.

   c. You are working for " COMPANY" that makes floats for ABC commodes. The floating ball has a specific gravity of 0.6 and has a radius of 5.5cm. You are asked to find the depth to which the ball is submerged when floating in water. The equation that gives the depth x to which the ball is submerged under water is given by 35

   $X^4-0.165\ X^2+3.993\text{x}10^{-2}=0$

   Use the false-position method of finding roots of equations to find the depth x to which the ball is submerged under water. Conduct three iterations to estimate the root of the above equation. Find the absolute relative approximate error at the end of each iteration, and the number of significant digits at least correct at the end of third iteration.

   d. The vertical distance in meters covered by a rocket from 8=t to 30=t seconds is given by 35

   $$s = \int_{8}^{30} \left(200 \ln\left[\frac{140000}{140000 - 2100t}\right] - 9.8t\right) dt$$

   Use Simpson 3/8 multiple segments rule with six segments to estimate the vertical distance. Plot also the curve and the data in tabular format.

   e. The vertical distance in meters covered by a rocket from 8=t to 30=t seconds is given by 35

   $$s = \int_{8}^{30} \left(200 \ln\left[\frac{140000}{140000 - 2100t}\right] - 9.8t\right) dt$$

   i. Use the single segment trapezoidal rule to find the distance covered for 8= t to 30=t seconds.
   ii. Find the true error, E_t for part (i).
   iii. Find the absolute relative true error for part (i).
   iv. Plot also the curve and the data in tabular format.

   f. How is the length of a bluegill fish related to its age? Use Polynomial regression to analyze this problem with below information. 35

   In 2023,n = 78 Rui fish were randomly sampled from PSTU Pond. The researchers measured and recorded the following data (**dataset**):

   - Response (y): length (in mm) of the fish
   - Potential predictor (x1): age (in years) of the fish

   The researchers were primarily interested in learning how the length of a Rui fish is related to its age.
2. Viva Voce 20
3. Project 10
4. Continuous lab assignment submission 5

Dept. of Computer and Communication Engineering
Faculty of Computer Science and Engineering
Patuakhali Science and Technology University
Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: I Session: 2018-2019

Course Code	**Course Title**	**January-June 2021**	**Credit:** 03
CCE-313	Computer Network		**Time:** 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 **[A.]** a) Describe the most popular wireless Internet access technologies today. Compare and contrast them. 2+2
b) List four access technologies. Classify each one as home access, enterprise access, or wide-area wireless access.
DSL, ADSL, Ethernet, 4G, 5G)

[B.] Write the duties/services and protocol of each specific OSI layer. 4

[C.] Consider an application that transmits data at a steady rate (for example, the sender generates an N-bit unit of data every k time units, where k is small and fixed). Also, when such an application starts, it will continue running for a relatively long period of time. Answer the following questions, briefly justifying your answer: 3

a. Would a packet-switched network or a circuit-switched network be more appropriate for this application? Why?

b. Suppose that a packet-switched network is used and the only traffic in this network comes from such applications as described above. Furthermore, assume that the sum of the application data rates is less than the capacities of each and every link.
Is some form of congestion control needed? Why?

[D.] Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates R1 = 500 kbps, R2 = 2 Mbps, and R3 = 1 Mbps. 3
a. Assuming no other traffic in the network, what is the throughput for the file transfer? — 500kb
b. Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?
c. Repeat (a) and (b), but now with R2 reduced to 100 kbps.

2 **[A.]** How are the blocks allocated? What are the restrictions need to be applied to the allocated block. 3

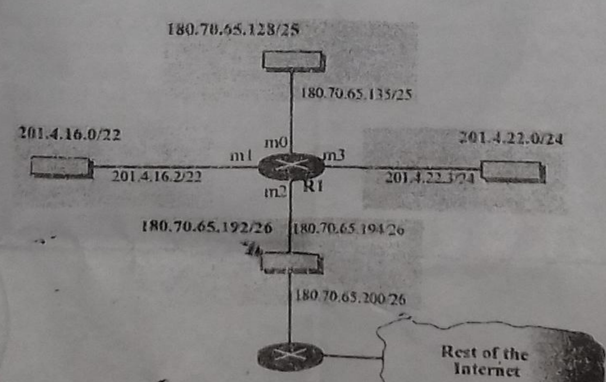
[B.] One of the addresses in a block is 110.23.120.14/20. Find the number of addresses, the first address, and the last address in the block. 3

[C.] An organization is granted a block of addresses with the beginning address 254.24.252.0/24. The organization needs to have 4 subblocks of addresses to use in its four subnets as shown below: 6
One subblock of 123 addresses.
One subblock of 62 addresses.
One subblock of 11 addresses.
One subblock of 5 addresses.

[D.] Write the steps need to be carefully followed to guarantee the proper operation of the subnetworks. 2

3 **[A.]** Discuss the packet forwarding without Subnetting. 3

[B.] Show the forwarding process if a packet arrives at R1 in below Figure with the destination address 180.70.65.140. 3

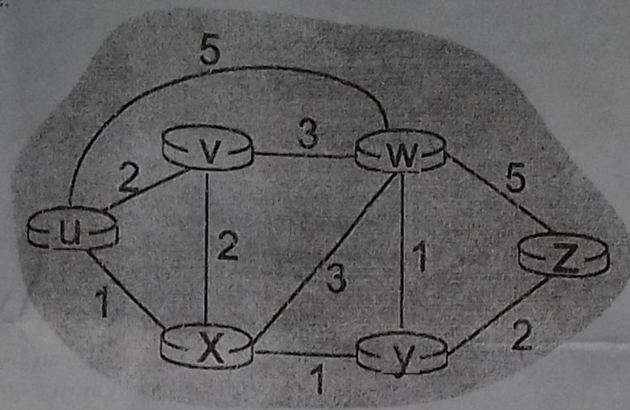


1011
0011
10001

[C.] Compare and contrast the properties of a centralized and a distributed routing algorithm. Give an example of a routing protocol that takes a centralized and a decentralized approach. 3

[D.] Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from u to all network nodes. Show how the algorithm works by computing a table. 5

TFT



4 [A.] The following is a dump of a UDP header in hexadecimal format. 5

0045DF000058FE20

a. What is the source port number?
b. What is the destination port number?
c. What is the total length of the user datagram?
d. What is the length of the data?
e. Is the packet directed from a client to a server or vice versa?
f. What is the client process?

[B.] How do opening, closing, and shrinking of window occur in TCP? Give examples for both send and receive window. 4

[C.] What are the services that a transport-layer protocol can offer to applications invoking it? Describe each with example. 3

[D.] A sender sends a JPEG message. Show the MIME header. 2

5 [A.] Suppose, An FTP server has received a packet from an FTP client with IP address 153.2.7.9. The FTP server wants to verify that the FTP client is an authorized client. The FTP server can consult a file containing the list of authorized clients. However, the file consists only of domain names. The FTP server has only the IP address of the requesting client, which was the source IP address in the received IP datagram. The FTP server asks the resolver (DNS client) to send an inverse query to a DNS server to ask for the name of the FTP client. Show the query and response messages with values for each field. Discuss separately. 5

[B.] Describe how Web caching can reduce the delay in receiving a requested object. Will Web caching reduce the delay for all objects requested by a user or for only some of the objects? Why? 4

[C.] What are the services provided by the SSH protocol through port forwarding? Explain with example. 3

[D.] Show the encoding for the INTEGER 1456 in case of SNMP. 2

6 [A.] Suppose a content provider, NetCinema, employs the third-party CDN company, KingCDN, to distribute its videos to its customers. On the NetCinema Web pages, each of its videos is assigned a URL that includes the string "video" and a unique identifier for the video itself; for example, *Transformers 7* might be assigned http://video.netcinema.com/6Y7B23V. If DASH is used, what will be the steps before the client can dynamically select chunks from the different versions of the video? 5

[B.] Consider a new peer Alice that joins BitTorrent without possessing any chunks. Without any chunks, she cannot become a top-four uploader for any of the other peers, since she has nothing to upload. How then will Alice get her first chunk? 4

[C.] Define Jitter. Explain the application of TFTP in conjunction with DHCP. 3

[D.] Do you think H.323 is actually the same as SIP? Make a comparison between the two. 2

TFTP

Patuakhali Science and Technology University

5th Semester (Level-3, Semester-I) Final Examination of B.Sc. Engg. (CSE) (January-June 2021)

Course Code: CIT-311 Course Title : Microprocessors and Assembly Language

Credit Hour : 3.00 Session: 2018-2019 Full Marks:70 Duration: 3 Hours

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]

*Answer any 5 of the following questions. Answer must be **brief, relevant and neat.***

1 a) i. How can you determine the microprocessor as 8-bit or 16-bit or 32 bit or 64 bit? 2
ii. Distinguish between 8085 and 8086. 2
b) i. Mention the role of segment register during protected mode operation. 2
ii. Distinguish microcontroller and microprocessor. 2
c) i. Shortly describe 82C55 PPI with its operational modes. 2.5
ii. Why is memory decoding necessary in computer system? 1.5
d) Criticize the statement "More registers integration produce faster CPU". 2

2 a) i. Compare PROM, EPROM, and EEPROM. 1.5
ii. Give the evolution of microprocessor from mechanical era to present (with important advancement). 2.5
b) i. How does the DMA speed up CPU performance? +1 2
ii. Distinguish between SRAM and DRAM. +1 2
c) i. Why is stepper motor so called? 1.5
ii. "All coprocessors are peripherals, but all peripherals are not coprocessors"-Explain this statement. 2.5
d) "CPU actually works on binary digits"- Explain this statement. 2

3 a) i. Suppose you would like to transfer data from your disk drive to a flash drive by using DMA controller. Explain whole procedure in details to complete the activities. 2
ii. Describe overlapping data movement mechanism of DMA. 2
b) i. Describe the responsibility of memory management unit in computer system. 3
ii. Calculate the number of page table entries that are needed for following combinations of virtual address size (n) and page size (P). 1

n	$P=2^p$	#PTE
16	4K	
64	16K	

c) i. Define handshaking. 1
ii. Explain page fault handling mechanism in virtual memory system. 3
d) "CPU actually works on binary digits"- Explain this statement. 2

4 a) Which Intel microprocessor addresses 1T of memory? Draw the block diagram of a computer system. [3]

b) What is a displacement? How does it determine the memory address in a MOV DS:[2000H],AL instruction? [2]

c) Explain the difference between the MOV BX, DATA instruction and the MOV BX, OFFSET DATA instruction. [2]

d) What, if anything, is wrong with a MOV AL,[BX][SI] instruction? Suppose that DS = 1200H, BX = 0100H, and SI = 0250H. Determine the address accessed by each of the following instructions, assuming real mode operation: [4]

i. MOV [100H],DL
ii. MOV [SI+100H],EAX
iii. MOV DL,[BX+100H]

c) How many bytes are stored on the stack by a PUSH AX? Show which JMP instruction assembles (short, near, or far) if the JMP THERE instruction is stored at memory address 10000H and the address of THERE is: [3]

i. 10020H
ii. 11000H
iii. 0FFFEH
iv. 30000H

5 a) Write down the formats of the 8086–Core2 instructions. (a) The 16-bit form and (b) the 32-bit form. [2]

b) The effect of the PUSH AX instruction on ESP and stack memory locations 37FFH and 37FEH.[Assume SS=0300, ESP= 0 7 F E] [2]

c) Draw a diagram and show the LDS BX,[DI] instruction loads register BX from addresses 11000H and 11001H and register DS from locations 11002H and 11003H. This instruction shows at the point just before DS changes to 3000H and BX changes to 127AH. The initial value of DS= 1 0 0 0 and EDI= 1 0 0 0. [3]

d) Convert machine code 8BEC to equivalent assembly instruction. [4]

e) If the start of a segment is identified with .DATA, what type of memory organization is in effect? What values appear in SP and SS if the stack is addressed at memory location 02200H? [3]

~~2000~~ +200

6 a) Define DAA and DAS. Show the process of addition with-carry, how the carry flag (C) links the two 16-bit additions into one 32-bit addition. [3]

b) What is wrong with the ADD RCX, AX instruction? If AX = 1001H and DX = 20FFH, list the sum and the contents of each flag register bit (C, A, S, Z, and O) after the ADD AX, DX instruction executes. [3]

c) What is the difference between the NOT and the NEG instruction? List the number of data items stored in each of the following memory devices and the number of bits in each datum: [3]

i. 2K × 4
ii. 1K × 1
iii. 4K × 8
iv. 16K × 1
v. 64K × 4

d) Which type of JMP instruction (short, near, or far) assembles for the following: [3]

i. if the distance is 0210H bytes
ii. if the distance is 0020H bytes
iii. if the distance is 10000H bytes

e) Contrast minimum and maximum mode 8086/8088 operation. Explain the operation of the pin. [2]

Patuakhali Science and Technology University
Department of Computer Science and Information Technology

Semester (Level-3, Semester-I), **Final Examination of B.Sc. Engg.(CSE)**, January-June/2021, Session 2018-19
Course Code: CIT-312 Course Title: Microprocessor and Assembly Language Sessional
Full Marks: 70 Duration: 3.00 Hours

[Figures in the right margin indicate full marks]

1802073

Answer all the following questions.

1. Use 8086 Emulator to write Assembly Language code solve the marked question

i	Write a program to print a message using individual letters and ASCII code of letters	20
ii	Write a program to add two numbers as well as form a Fibonacci series.	20
iii	Write a program to exchange the value of AX and BX.	20
iv	Write a program to Find Square Root of a number	20
v	Write a program to print a String	20
vi	Write a program to subtract two 8 bit BCD numbers	20
vii	Write a program to multiply two 16-bit numbers	20
viii	Write a program to subtract two 16-bit numbers with or without borrow	20
ix	Write a program to add two 8 bit BCD numbers	20
x	Write a program for Binary To Decimal Conversion	20
xi	Write a program to find the factorial of a number	20
xii	Write a program for Decimal to Binary Conversion	20
xiii	Write a program to add two 16 bit numbers	20

2. Microcontroller Based System Design Project 30

3. Viva Voce 20

same A-M

Patuakhali Science and Technology University

5th Semester (Level-3, Semester-I) Final Examination of B.Sc. Engg. (CSE) (January-June 2021)

Course Code: **CIT-313** Course Title : **Computer Organization and Architecture**

Credit Hour : 3.00 Session: 2018-2019 Full Marks:70 Duration: 3 Hours

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions. Answer must be ***brief, relevant and neat.***

1	a) Describe RAID working mechanism.	3
	b) "If a large number of device are connected to the buses, performance will suffer"- explain this statement. Distinguish among local bus, system bus and expansion bus.	4
	c) Give the characteristics of RISC and CISC.	3
	d) Describe about set associative mapping. Give the replacement algorithms in cache memory.	4
2	a) Shortly explain the working mechanism of operating system to interact user with computer hardware.	3
	b) Find out the quotient and reminder by performing twos complement division operation where divisor = -3 and dividend = -7.	4
	c) Define random access memory. Distinguish between two's complement and sign magnitude representation.	3
	d) Consider a machine with a byte addressable main memory of 2^{16} bytes and block size of 8 bytes. Assume that a direct mapped cache consisting of 32 lines is used with this machine.	
	i) How is a 16-bit memory address divided into tag, line number, and byte number?	1
	ii) Into what line would bytes with each of the following addresses be stored? i. 0001 0001 0001 1011 ii. 1100 0011 0011 0100	2
	iii) How many total bytes of memory can be stored in the cache?	1
3	a) Distinguish between program driven and interrupt driven I/O.	3
	b) i) Illustrate and list the micro-operations to perform addition between two operands from memory.	4
	ii) Assume numbers are represented in 6-bit two's complemented representation. Show calculation of the following. -1A+2B (Hex)	
	c) Why I/O module is used in computer system? Classify the external devices.	3
	d) Write short notes (any two) i) Dual core. ii) Pentium 4. iii) Core i7.	4
4	a) What, in general terms, is the distinction between computer organization and computer architecture? List and briefly define the main structural components of a computer.	3
	b) Explain Moore's law. On the IAS, describe the process that the CPU must undertake to read a value from memory and to write a value to memory in terms of what is put into the MAR, MBR, address bus, data bus, and control bus.	4
	c) What are the key properties of semiconductor memory? What is the difference between DRAM and SRAM in terms of characteristics such as speed, size, and cost?	3
	d) Define memory cell. Describe the structure of a typical Memory Cell.	4
5	a) What is chip logic? Describe a 16 Megabit DRAM (4M × 4) memory structure.	3
	b) How Error-Correcting Code works? Describe the Hamming Error-Correcting Code procedure.	4
	c) Write down the elements of a machine instruction. Draw and describe the Instruction Cycle State Diagram.	3
	d) Describe about Nested Procedures. What is the difference between an arithmetic shift and a logical shift?	4
6	a) Define register addressing. Describe the x86 Instruction Formats.	3
	b) What is user-visible registers? Draw the Internal Structure of the CPU.	4
	c) Write down the control and status registers. Describe the data flow of fetch and indirect cycle.	3
	d) What is instruction pipelining? Describe the Six-Stage CPU Instruction Pipelining with timing diagram.	4

2 8 6 511 5 0011 0111 111001011 2 111001 change restore 0 15001 101011 100110 010001 0111 1000 1100

FDEXX S

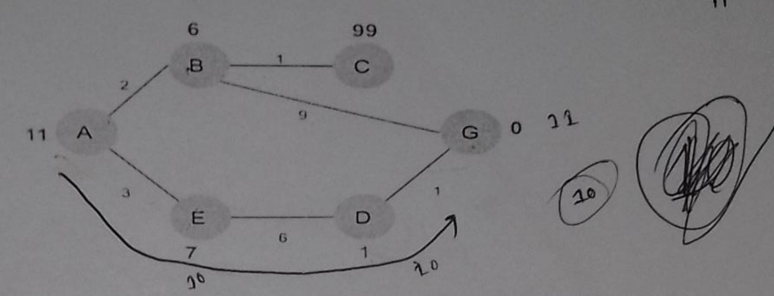
Patuakhali Science and Technology University

B.Sc. Engg.(CSE) 5th Semester (Level-3 Semester-I) Final **Sessional Examination** of January-June 2021
Course Code: CIT-315 Course Title: Artificial Intelligence Sessional
Session 2018-19 Credit Hour: 1.5 Full Marks: 70 Duration: 2.50 Hours.

Implement one from each Lab work.

Lab01 Given the below graph, numbers written on edges represents the distance between nodes and numbers written on nodes represent the heuristic value. Implement the A star search algorithm for estimating the cost-effective path from A to G, where A is assigned as source node and G is assigned as the goal node. 10

8 202 202 11



Implement the Hill-climbing search algorithm for solving the following slide puzzle.

3	8	5
	7	1
2	6	4

(a). Puzzle initial state.

1	2	3
4	5	6
7	8	

(b). Puzzle final state.

Lab02 Implement the Naïve bayes classifier for checking weather to play or not on weather data. 10

Outlook	Temperature	Humidity	Windy	Play
sunny	hot	high	false	no
sunny	hot	high	true	no
overcast	hot	high	false	yes
rainy	mild	high	false	yes
rainy	cool	normal	false	yes
rainy	cool	normal	true	no
overcast	cool	normal	true	yes
sunny	mild	high	false	no
sunny	cool	normal	false	yes
rainy	mild	normal	false	yes
sunny	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
rainy	mild	high	true	no

Build a Decision Tree based on the above dataset. Test the new case with your decision tree.

Lab03 See Attachment 10

Lab04 AI project on ChatBot. 20

5. List the given problems which have been solved by you in the sessional classes. 10

6. Viva-voce. 10

Sun

5 - No $\frac{5}{14}$

9 - Yes = $\frac{9}{14}$

	Y	No
Suny	$\frac{3}{9}$	$\frac{}{5}$